Convergence and Accuracy of PINNs vs. Classical Finite Difference Schemes for BVPs

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Outline

1. Motivation
2. Prior Work
3. Test Problems
4. Finite Difference Methods
5. Nonlinear Solvers (Problem 3)
6. Physics-Informed Neural Networks
7. Results
8. Conclusions

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Research question: *How do PINNs compare to classical finite difference methods in terms of convergence and accuracy for solving boundary value problems?*

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Another note: **Spectral Bias**, Rahaman et al., (2019)

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$$(1 + x^2)\, u'' + 2x\, u' = d(x), \quad u(0) = 1,\ u(1) = 2 + \sin(1)$$

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Problem 3 (nonlinear):

$$u'' = 3u + x^2 + 10u^3, \quad u(0) = 0,\ u(1) = 0.1$$

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- Condition number: Predictable accuracy ceiling

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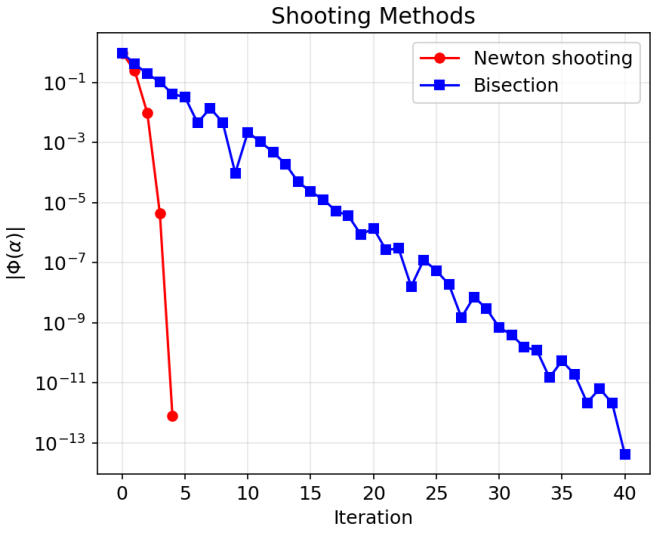

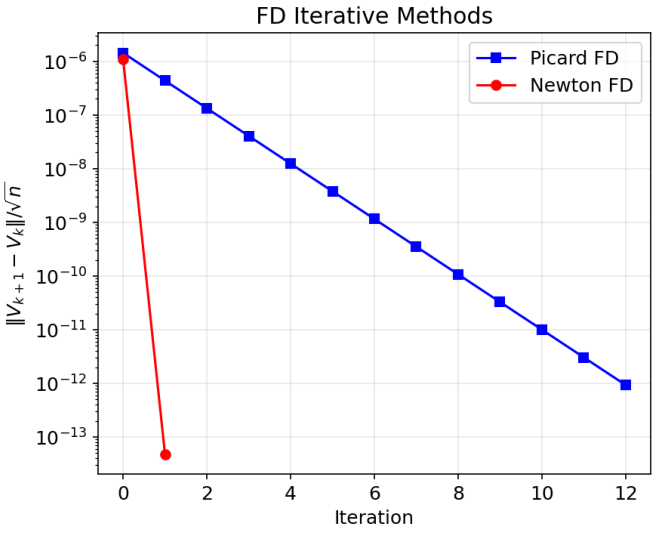

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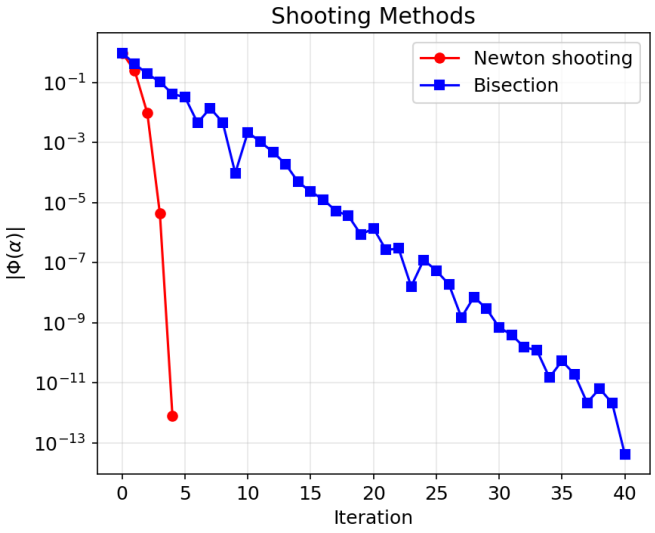

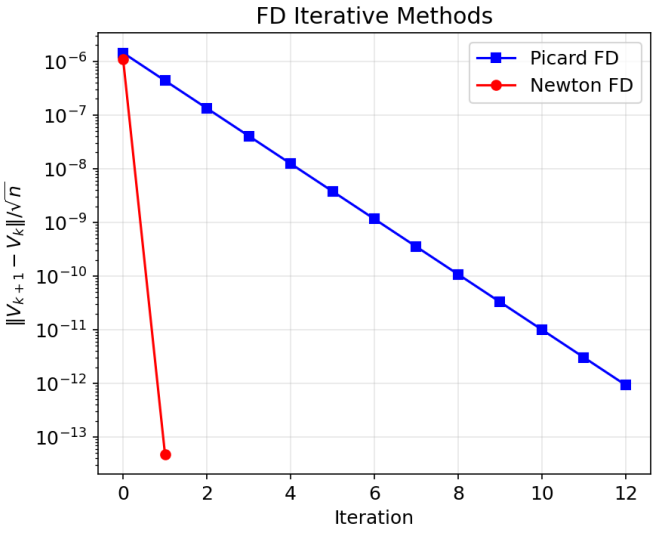

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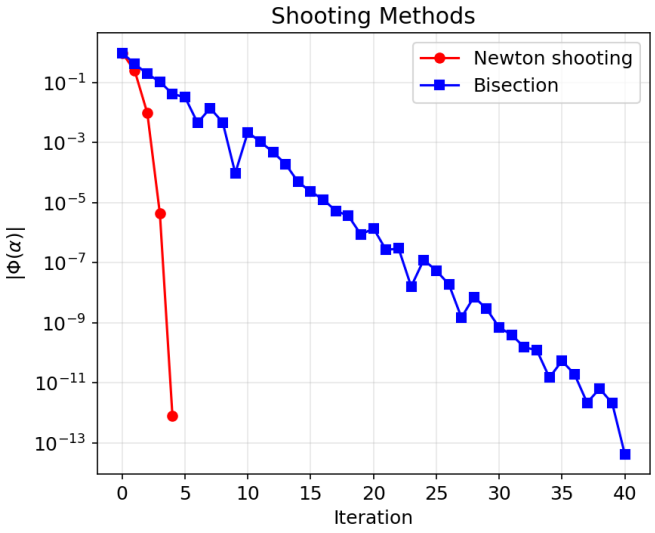

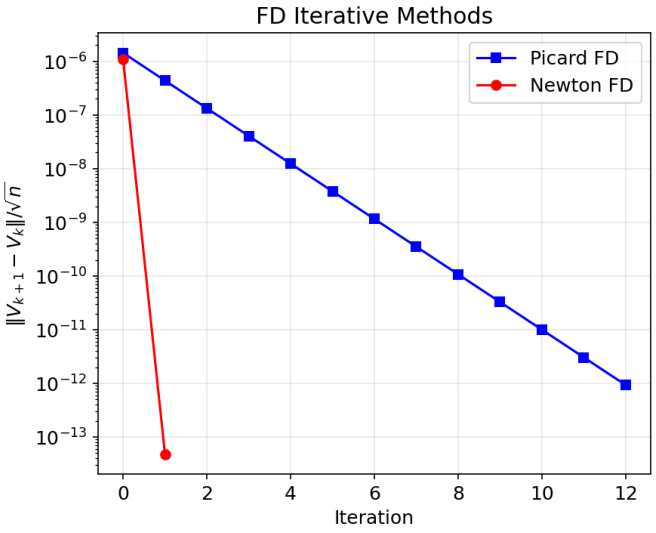

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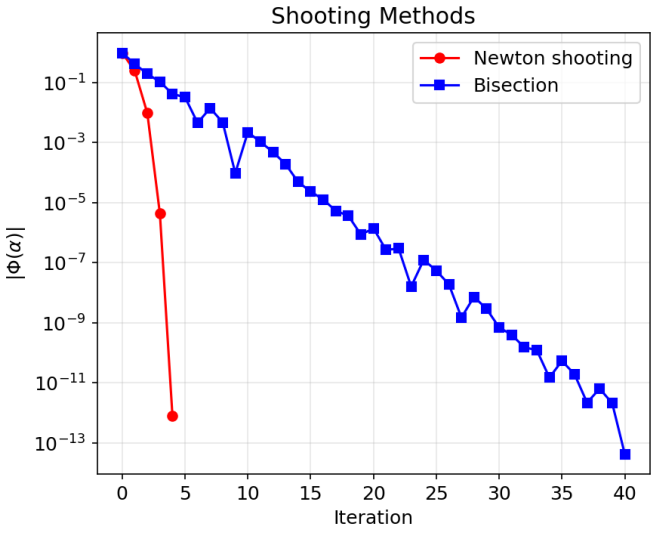

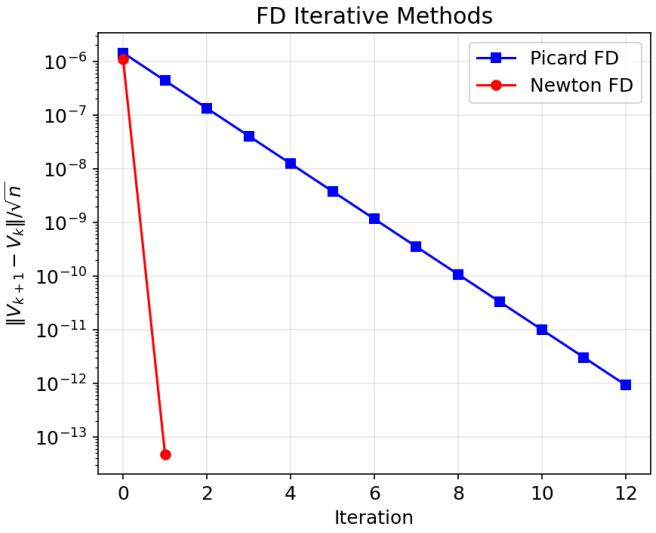

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- Architecture: 3 hidden layers $\times$ 50 neurons, $\tanh$, $\approx$ 3,200 parameters

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- Increasing N_c does not help: the bottleneck is the optimizer, not the sampling

Results: FDM Convergence

	Problem 1		**Problem 2**		**Problem 3**	
n	Error	Rate	Error	Rate	Error	Rate
7	1.26e-2	–	3.91e-3	–	3.75e-4	–
31	7.79e-4	1.92	2.45e-4	1.91	2.39e-5	1.91
127	4.86e-5	1.96	1.53e-5	1.98	1.49e-6	1.98
511	3.07e-6	1.93	9.59e-7	1.99	9.33e-8	1.99

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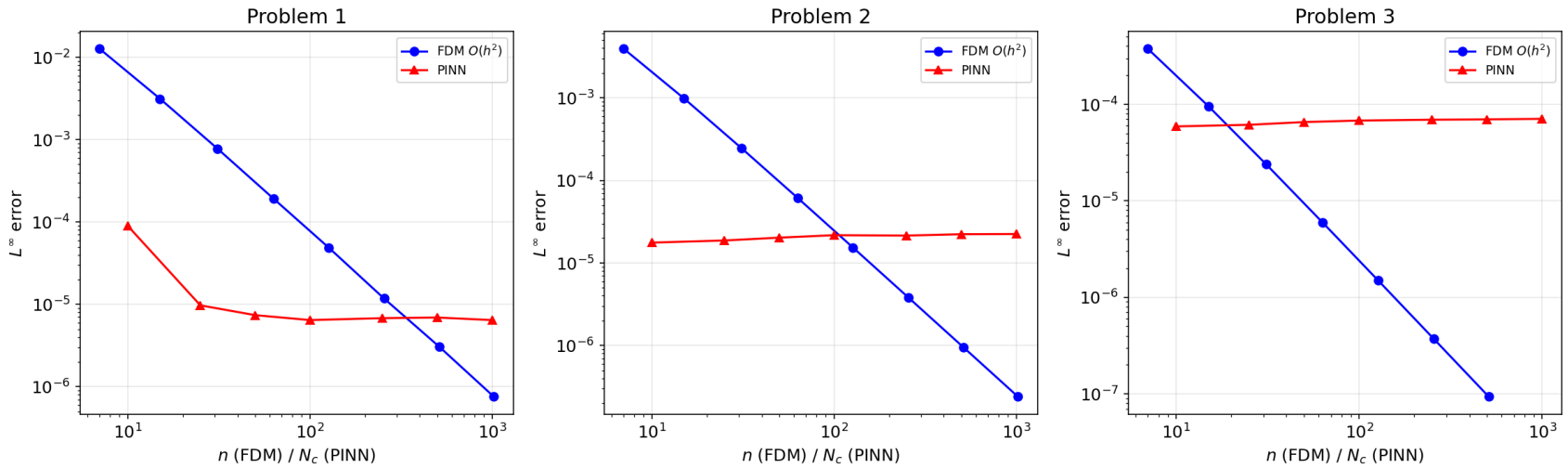
- Problem 3 converges identically despite nonlinearity

Results: Accuracy vs. Sampling Density



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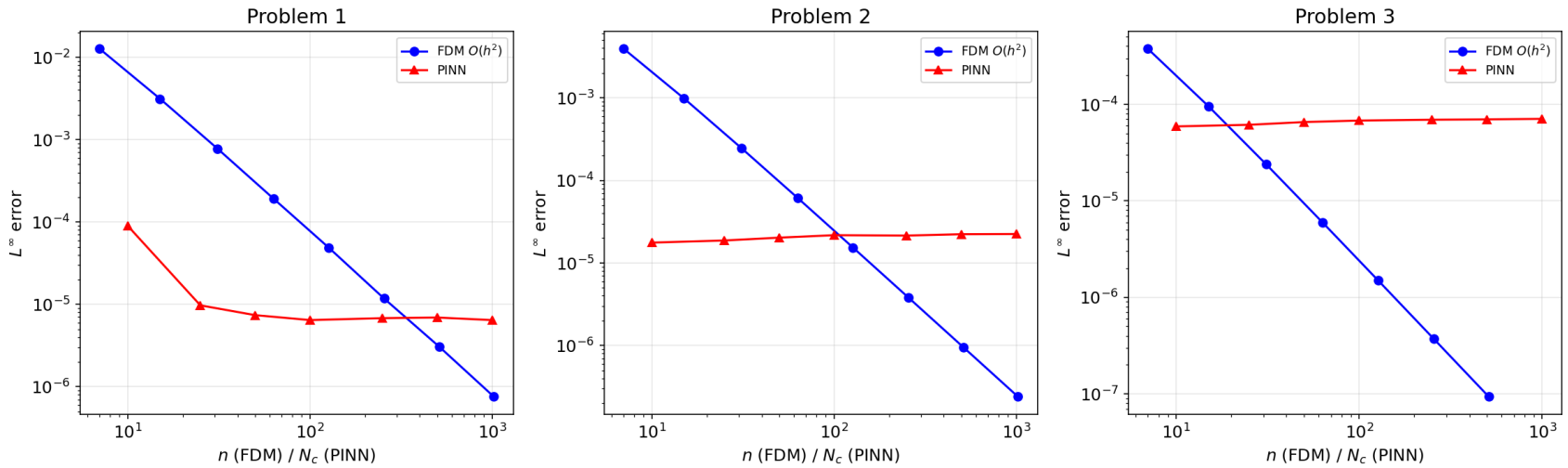

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- PINN (red): **flat**. Increasing N_c does not reduce error.

Results: PINN Plateau

N_c	**Problem 1**	**Problem 2**	**Problem 3**
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50	7.39e-6	2.03e-5	6.55e-5
100	6.44e-6	2.17e-5	6.79e-5
500	6.91e-6	2.23e-5	6.98e-5
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- Compare: FDM at $n = 511$ achieves 9.33e-8 (Problem 3)

Results: Cost

Wall-Clock Time

Method	Time
FDM ($n = 255$)	$\sim 200\,\mu$s
FDM ($n = 1023$)	$\sim 370\,\mu$s
PINN training	41–70 s
PINN inference	$\sim 400\,\mu$s

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- PINN inference (post-training) $\approx$ FDM solve time

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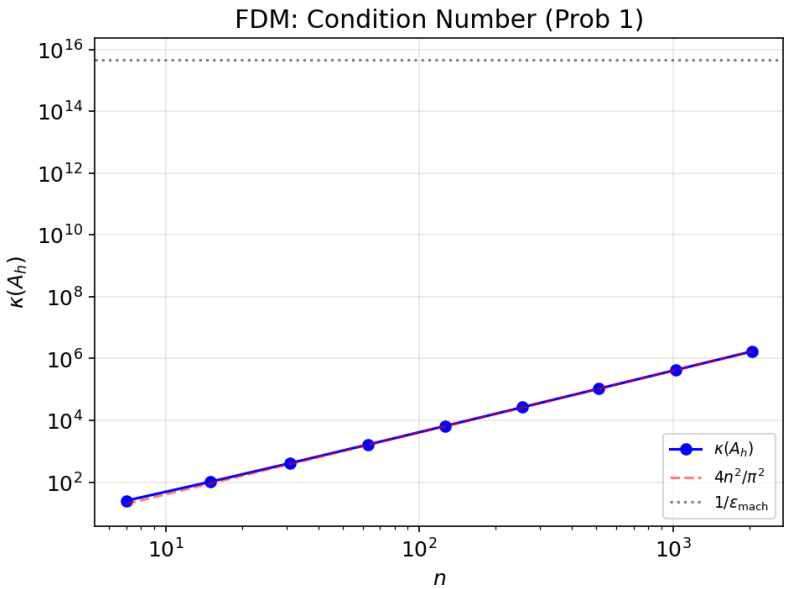

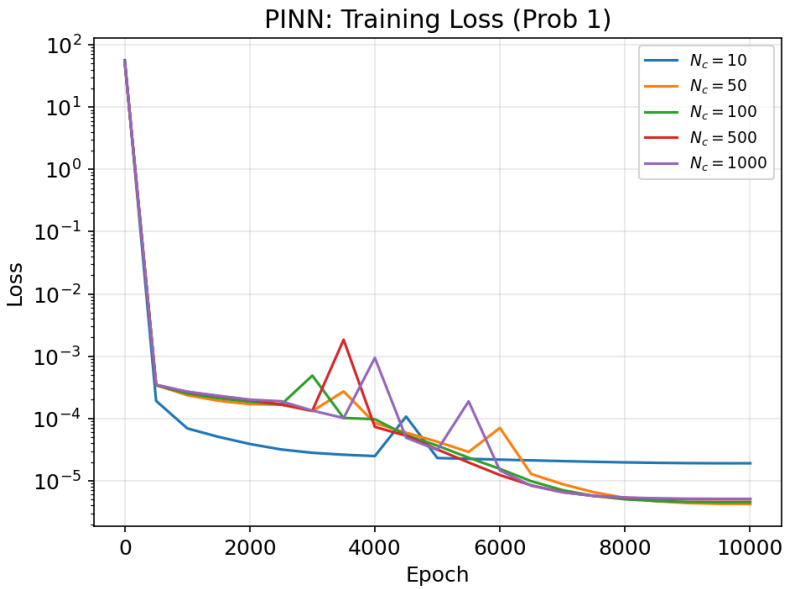

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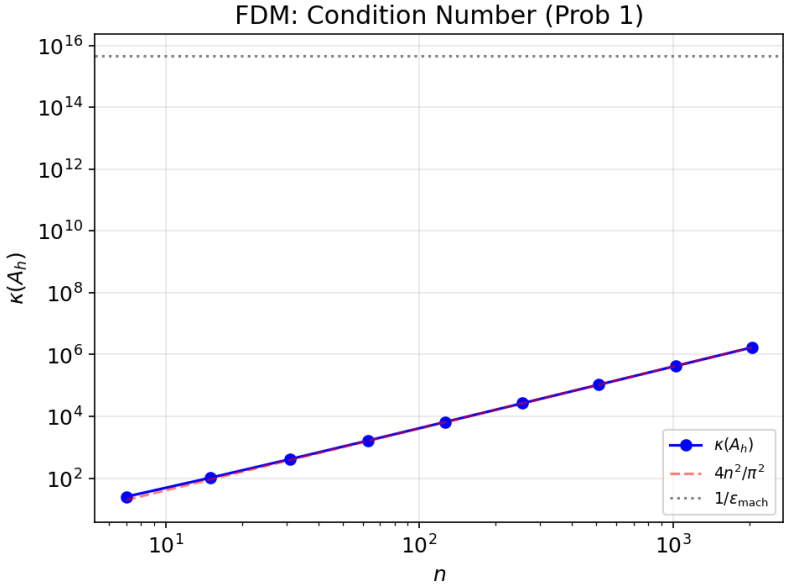

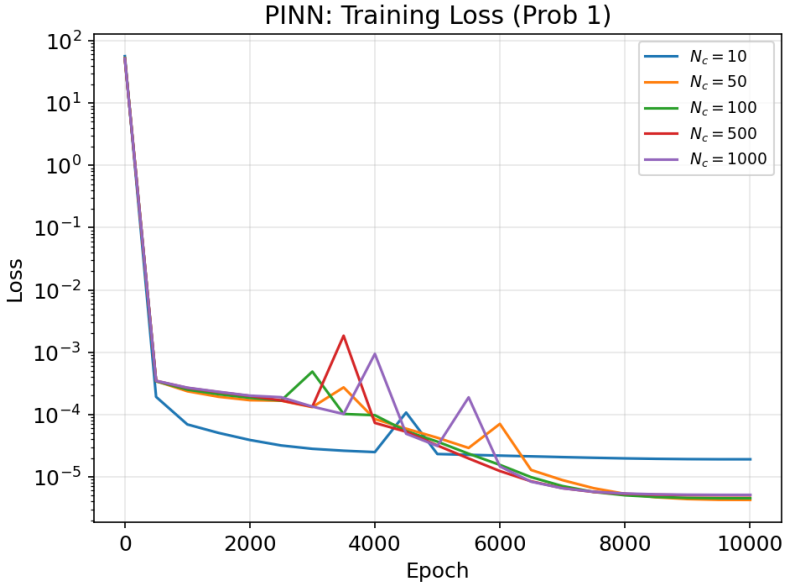

- FDM ceiling: predictable ($\kappa \cdot \epsilon_{\text{mach}}$)
- PINN ceiling: empirical, no theory to predict it

Conclusions

	FDM	PINN
Convergence rate	$O(h^2)$ (proven)	No guarantee
Best accuracy	$\sim 10^{-7}$ (limited by κ)	$\sim 10^{-5}$ (plateau)
Wall-clock time	μs	13–78 s (training)
Mesh required?	Yes	No
Nonlinear problems	Iterative solvers	Same framework
Inverse problems	No mechanism	Natural fit

- FDM is superior for well-posed 1D BVPs
- PINN accuracy plateau consistent with Wang et al. (2022)
- PINNs may shine in higher dimensions, complex geometries, inverse problems... further research?

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